



SMART ROBOTS Working Collaboratively With Humans



An industrial engineer using software to monitor machine in real time

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Scientists all over the world are developing intelligent robots that can observe and learn from experience and human behaviour. The robots can then collaborate with humans across a range of sections

As technology advances, the range of potential applications for robots grows even wider. Robots are becoming more flexible, accurate, safer and faster. These are more energy efficient, compact and lighter. Modern robots are easier to program and capable of integrating with a wider range of machines.

Use of industrial robots has increased steadily in recent years in India. Along with the cloud, these play a key role in production, as the most flexible elements. Robotic automation is indispensable for anyone wishing to rise to the challenges of Industry 4.0.

Amit Maheshwari, technical lead - robotics and automation, Ingen Dynamics Inc., says, "Latest technology in robotics is integrating robots with artificial intelligence (AI), so that these can interact with humans in a better way. This is done by making the robots understand emotions, faces and expressions of people. The robots could then replicate the same, and interact in ways similar to having a normal conversation. We would not know over a phone call if the other side is a human or robot.

"Personal robots is a recent application

area. Warehouse automation is another. Robots like unmanned vehicles are being used in the defence sector. Exploration is going on with underwater robots. Next stage of evolution is more flexible industrial robots."

Achu Wilson, co-founder, Sastra Robotics, explains, "What separates smart robots from traditional industrial automation is the capability of the robots to adapt to various situations and work as intended. This capability is mainly brought in by advances in AI, machine learning and sensors, among others.

"Traditional robots work excellently in well-defined, controlled factory or production environments. These do not have sensing or decision-making capabilities to operate in regular, cluttered human environments. For example, robot manipulators are used in factories to pick/place/process thousands of similar items. But these cannot perform well with other objects.

"Amazon's picking challenge aims to develop such intelligent capabilities for robot arms that enable these to pick up thousands of varying objects in a typical e-commerce warehouse. These sensing and decision-